

Superposition: Net Force from Three Charges

Coulomb Forces on Three Charged Objects · oPhysics

Senior Secondary (Years 11-12)

~30 minutes

Science · Physics

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Simulation: <https://ophysics.com/em1a.html>

Curriculum links: electric fields · electricity

Learning intentions

- I can apply the principle of superposition to combine electric forces from multiple charges.
- I can add two force vectors to predict a net force direction and compare it to a simulation.

Getting started

Open the simulation and place three charges in a straight line, with the object you are analysing in the middle. Note the sign and magnitude of each charge.

Tasks

1. With three charges in a line (all positive), calculate the net force on the middle charge by hand, then compare to the simulation's displayed force vector.

Working:

2. Rearrange the three charges into a triangle. Sketch the two individual force vectors on the object you are analysing, then sketch your predicted net force vector before checking the simulation.
3. Change the sign of one of the outer charges and predict how the net force direction will change. Check your prediction.
4. Set up three different charge configurations and record the magnitude and direction of the net force for each.

Configuration

Net force magnitude

Net force direction

Extension (optional)

Explain how the principle of superposition demonstrated here also explains the shape of the electric field around complex charge distributions, such as a charged rod.

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